Original Article

Corrosion Inhibition Studies of *Dioscorea Bulbifera* Leaves Extract on Aluminum Metal in 0.6 M Hydrochloric Acid Solution

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ABSTRACT

This study evaluates the corrosion inhibition performance of *Dioscorea bulbifera* leaf extract on aluminum in 0.6 M HCl using weight loss (ΔW) measurements at 303–323 K over 4 hours. The results show that increasing extract concentration (0.0–0.6 g) significantly reduces ΔW and corrosion rate (CR) at all temperatures. At 303 K, CR decreased from 0.035 to 0.029, while inhibition efficiency (%IE) increased slightly from 62% to approximately 63–64%, indicating effective surface protection. A similar trend was observed at 313 K and 323 K, where CR decreased with concentration, accompanied by slight improvements in %IE. Surface coverage (θ) also increased marginally, confirming enhanced adsorption of inhibitor molecules on the aluminum surface. However, temperature had a pronounced effect on inhibition performance. As temperature increased from 303 to 323 K, CR increased while %IE decreased from 62 to 43% at the blank concentration, indicating reduced inhibitor efficiency. This decline is attributed to thermal desorption of adsorbed molecules at higher temperatures. The behavior suggests that the inhibition mechanism is predominantly governed by physisorption, involving weak intermolecular interactions. Thus, *Dioscorea bulbifera* extract exhibits moderate %IE, with optimal performance at lower temperatures.

Introduction

Corrosion, defined as the gradual deterioration of metals through chemical or electrochemical interaction with their environment, remains a critical challenge in industrial and engineering applications [1-3]. Aluminum, despite its widespread use due to its low density, high conductivity, and natural oxide film that provides some degree of passivation, is highly susceptible to corrosion in aggressive acidic environments such as hydrochloric acid (HCl) solutions [4]. HCl is extensively used in industrial processes including pickling, acid cleaning, descaling, and oil well acidizing, where aluminum components are often exposed to highly corrosive conditions [5]. In such media, the protective oxide layer on aluminum dissolves, leading to accelerated metal degradation, reduced efficiency, and increased economic losses [6]. Therefore, the development

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of effective corrosion inhibition strategies is essential for prolonging the lifespan of aluminum materials and ensuring operational safety [7]. Corrosion inhibitors are substances that, when added in small concentrations to a corrosive medium, significantly reduce the CR of metals by adsorbing onto the metal surface and forming a protective barrier [8]. Traditionally, inorganic inhibitors such as chromates, phosphates, and nitrites have been widely employed due to their high efficiency. However, these inhibitors pose serious environmental and health concerns due to their toxicity, non-biodegradability, and potential for bioaccumulation [9]. Increasing environmental regulations and the global shift toward sustainable chemistry have necessitated the exploration of eco-friendly alternatives, leading to the emergence of green corrosion inhibitors derived from plant extracts [10]. Plant-based inhibitors have gained considerable attention because they are renewable, biodegradable, readily available, and environmentally benign. These natural inhibitors typically contain phytochemicals such as alkaloids, tannins, flavonoids, saponins, and phenolic compounds, which possess heteroatoms (such as nitrogen, oxygen, and sulfur) and π-electron systems that facilitate adsorption onto metal surfaces [11,12]. This adsorption forms a protective film that blocks active corrosion sites and reduces both anodic and cathodic reactions. Among various plant species investigated, *Dioscorea bulbifera (DB)*, commonly known as air potato, has shown promising potential due to its rich phytochemical composition and availability in tropical regions. Despite numerous studies on green inhibitors for aluminum corrosion in acidic media, several gaps remain in the literature. Many previous studies have focused on lower acid concentrations (typically 0.1–0.5 M HCl), leaving limited understanding of inhibitor performance under more aggressive conditions such as 0.6 M HCl, which are more representative of industrial applications. Additionally, earlier investigations focus only on the surface morphology and calculations such as surface coverage and weight loss only. Without providing the kinetics mechanism of inhibition, there is also insufficient emphasis on correlating surface morphology with inhibition performance, particularly using advanced characterization techniques such as scanning electron microscopy (SEM). Furthermore, limited attention has been given to the thermodynamic aspects of adsorption, including parameters such as activation energy, enthalpy, and entropy changes, which are crucial for understanding the nature (physisorption or chemisorption) and stability of the inhibitor film. Another notable gap is the lack of systematic studies examining the combined effects of inhibitor concentration and exposure time on corrosion behavior, which are essential for practical application and optimization. In light of these gaps, this study aims to provide a comprehensive evaluation of the corrosion inhibition performance of *DB* leaf extract on aluminum in 0.6 M HCl solution. The objectives of the study are to determine the %IE of the extract using ΔW measurements, to analyze the surface morphology of aluminum before and after inhibition using SEM, and to evaluate the thermodynamic parameters governing the adsorption process. Additionally, the study seeks to examine the effects of varying inhibitor concentration and immersion time on the CR and %IE, thereby providing a detailed understanding of the kinetics and mechanism of inhibition. The scope of this research is limited to laboratory-scale investigations using aluminum coupons immersed in 0.6 M HCl solution with varying concentrations of *DB* leaf extract. The study integrates gravimetric (ΔW), surface characterization, and thermodynamic analyses to provide a multidimensional assessment of inhibitor performance. While the findings are expected to contribute valuable insights into the development of green corrosion inhibitors, the study does not extend to industrial-scale applications or long-term field testing. The significance of this research lies in its contribution to the advancement of sustainable corrosion control strategies. By exploring the effectiveness of a plant-based inhibitor, this study supports the transition from toxic inorganic inhibitors to environmentally friendly alternatives. Green inhibitors provide several advantages, including low toxicity, biodegradability, cost-effectiveness, and minimal environmental impact, making them suitable for large-scale industrial use.

Ultimately, this work aims to enhance the understanding of corrosion inhibition mechanisms and promote the adoption of eco-friendly materials in corrosion management practices.

Experimental

Plant Extraction

The plant *DB* leaves were obtained at the BUK agro plant new site located in the Gwale Local Government Area of Kano State. The plant leaves were collected in accordance with analytical methods, where they were dried under the shed [11]. Ethanol (99% purity) was used as a solvent to extract *DB* leaves. 500 g of the ground powder leaves were used in 2 L of ethanol at a time with intermittent shaking for two weeks and filtered using Whatman filter paper [12]. The solution was evaporated using a Rotary evaporator, Model R-210, at 48 °C to obtain the concentrate crude extract. The crude extract was dried, stored in a clean bottle and covered properly for further experiments.

Preparation of Inhibitor Solution in 0.3 M HCl

To carry out the experiment, 0.2, 0.4, and 0.6 g L^{-1} concentrations of the extract were injected into 0.3 M HCl which was prepared by weighing 0.2, 0.4, and 0.6 g of the extract and dissolving each in 5 mL of ethanol. Then each was transferred into a 1 litre volumetric flask separately and made to the mark using 0.3 M HCl solution [13]. The solution of the co-respondent was made using Equation 1.

$$C_1V_1=C_2V_2 \quad (1)$$

Where, C_1 is the cncentration of stock solution, V_1 denotes volume of stock solution to be measured, C_2 represents the concentration to be prepared, and V_2 is the volume to be prepared.

Preparation of the Aluminum Coupons

The aluminum sheet used for this study was obtained from the Kofar Ruwa Market, Kano, Nigeria with dimensions of 1 ×1 × 0.11 mm. The sheet was branded and polished mechanically with silicon carbide emery paper (120, 400, 800, and 1,000 grade) for a smoother surface, degreased by washing in ethanol, dried in acetone, and stored in a desiccator.

Weight Loss Experiment

The coupons were immersed in 0.6 M HCl without and with inhibitors for 4 hours and washed at every hour to be able to determine the ΔW of the coupons during the period. The weight before immersion and after was determined. This test was conducted under total immersion conditions in 100 mL at 303, 313, and 333 K, respectively. The retrieved coupons were appropriately cleaned, dried, and reweighed. The ΔW was taken to be the difference between the weight of the coupons at a given time and their initial weight. All tests were run in triplicates and the data were collected. Average values for each experiment were used in subsequent calculations [14].
From the initial and final weights of mild steel, the ΔW, CR (g $h^{-1}cm^{-2}$), %IE, and the degree of surface coverage (θ) were calculated as shown in Equations 2-4.

$$\%IE = \left(1 - \frac{W1}{W2}\right)\times 100 \quad (2)$$

$$\theta = 1 - \frac{W1}{W2} \quad (3)$$

$$CR = \frac{\Delta W}{At} \quad (4)$$

Where, W1 and W2 are the ΔW (g) for mild steel in the presence and absence of *Sarcocephalus latifolius* extract respectively, θ is the degree of surface coverage of the inhibitor, A denotes the area of the metal coupon (in cm^2), t represents the time taken for the corrosion in hours, and ΔW denotes the W2 - W1, is the ΔW of mild steel after period, t [1].

Thermodynamic Calculations

Enthalpy is the sum of the internal energy of the system plus the product of the pressure of the gas in the system and its volume while entropy is the measure of disorderliness of a system [15]. The thermodynamic parameters for the enthalpy of adsorption and entropy of the inhibitors on the (ΔH_{ads}) and (ΔS_{ads}) in the presence and absence of inhibitors were

calculated using the transition state Equation 5 [16].

$$\frac{CR}{T} = \frac{R}{Nh} + \exp\frac{\Delta S_{ads}}{2.303R} = \exp\frac{-\Delta H_{ads}}{2.303RT} \quad (5)$$

Where, N is the Avogadro's number (6.02252×10^{23} mol^{-1}) and h is the Planck's constant (6.626176×10^{-34} Js), while other parameters have their usual definitions. A plot of ln $(\frac{C_R}{T})$ versus I/T gave a straight line with a slope equal to $(\frac{-\Delta H_{ads}}{2.303RT})$ and an intercept of ln $(\frac{R}{Nh}) + (\frac{\Delta S_{ads}}{2.303R})$ from Langmuir adsorption constant (K_{ads}) of *DB* is related to the free energy of adsorption according to Equations 3 and 4 [17].
The free energy of adsorption (ΔG_{ads}) was calculated using Equation 6:

$$\Delta G_{ads} = -RTIn(55.5\ K_{ads}) \quad (6)$$

Where, (K_{ads}) is the adsorption equilibrium constant obtained from adsorption isotherm models. These calculations provided insight into the nature and spontaneity of the inhibition process.

Kinetics Calculation

The kinetics of aluminum corrosion in HCl solution was studied by fitting data obtained from ΔW experiment into different kinetic equations of zero first and second orders, as derived by Equations 7-11.

$$\frac{dx}{dt} = k_1 (a_o - x) \quad (7)$$

Rearrangement of the Equation 7 produces Equation 8, and upon integration of Equation 8 yields Equations 9 and 10, respectively. From the Equations derived k_1 is the first order rate constant.

$$dx/(a_o - x) = k_1\, dt \quad (8)$$

$$-ln(a_o - x) = k_1\, t \quad (9)$$

$$-log\ (\Delta w) = \frac{k_1\, t}{2.303} \quad (10)$$

The half-life equation is obtained by fitting the value of the k_1 constant into the half-life equation, as shown in Equation 11.

$$t_{1/2} = \frac{0.693}{k_1} \quad (11)$$

A plot of –log (Δw) on the vertical axis against time, t at the horizontal axis, a linear graph was obtained and the values of the kinetic parameters were obtained from the graphs.

Results and Discussion

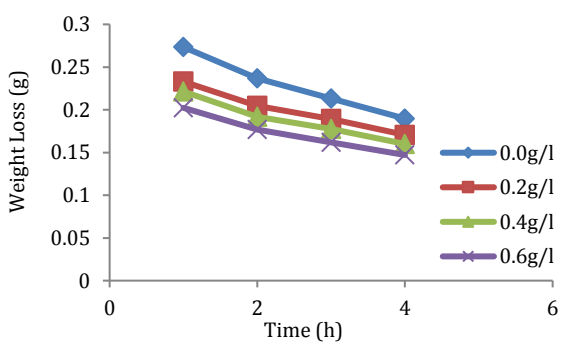
Weight Loss Experimental Result

The data in Table 1 describe the effect of increasing concentrations of *DB* extract and temperature (303–323 K) on aluminum corrosion in acidic medium. At 303 K, the ΔW and CR decrease as the extract concentration increases from 0.0 to 0.6 g. CR drops from 0.035 to 0.029, while %IE increases slightly from 62% to about 63–64%. This indicates that the extract acts as an effective corrosion inhibitor, forming a protective film on the aluminum surface [1,2,10]. The increase in surface coverage (θ ≈ 0.62 to 0.64) supports this, showing that more inhibitor molecules are adsorbing onto the metal surface with increasing concentration. At 313 K, a similar trend is observed: CR decreases (0.052 to 0.043) and %IE improves slightly (55 to 57%) with increasing concentration. However, the %IE is lower than at 303 K, suggesting that increasing temperature reduces inhibitor performance. This is likely due to desorption of inhibitor molecules from the aluminum surface at higher thermal energy. At 323 K, the same pattern continues: CR decreases with concentration (0.062 to 0.052), and %IE increases slightly (43 to 44%). However, overall efficiency is significantly lower than at 303 K and 313 K. This further confirms that inhibitor effectiveness decreases with temperature. Comparing across temperatures, for any fixed concentration, CR increases with temperature from 0.035 to 0.062 at blank concentration of the inhibitor in the same pattern, %IE decreases from 62 to 43% showing the effect of temperature in the corrosion process. This behavior suggests that the adsorption

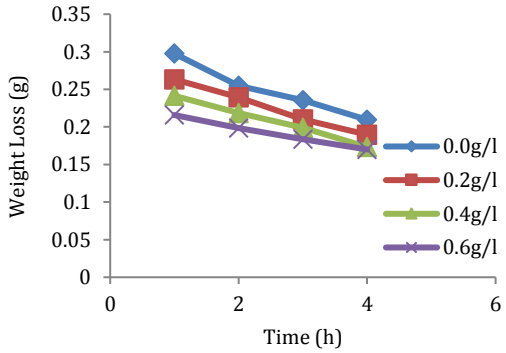
mechanism is predominantly physical adsorption (physisorption) rather than chemisorption [18]. In physisorption, the interaction between inhibitor molecules and metal surface is weak (van der Waals forces), thereby increasing temperature causes desorption, and reduces protection [14]. The effect of temperature on the concentration of acid is shown in Figures 1 and 2. Figure 1 depicts the trend of temperature on the weight loss across inhibitor concentrations in 0.6 M HCl of the acid over time.

Table 1: Δw, CR, surface coverage (Θ), and %IE of various concentration of *DB* in 0.6 M HCl solutions for 4 h

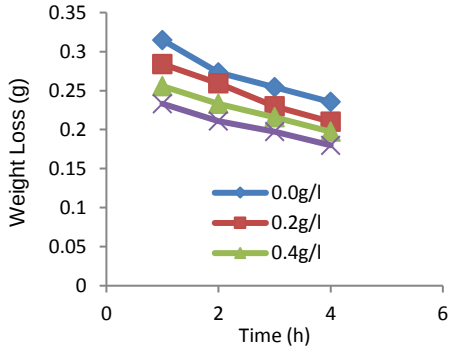
Conc. of extract (g/L)	303 K				313 K				323 K			
	ΔW(g)	CR	Θ	%IE	ΔW(g)	CR	Θ	%IE	ΔW(g)	CR	Θ	%IE
0.0	1.489	0.035	0.620	62	1.253	0.052	0.556	55	0.841	0.062	0.435	43
0.2	1.430	0.033	0.633	63	1.208	0.050	0.572	57	0.787	0.059	0.439	43
0.4	1.344	0.030	0.640	64	1.121	0.047	0.572	57	0.729	0.056	0.443	44
0.6	1.245	0.029	0.638	63	1.032	0.043	0.573	57	0.668	0.052	0.448	44



Effect of 303 K on 0.2 g inhibitor



Effect of 313 K on inhibitor



Effect of 323 K on inhibitor

Figure 1: Variation of ΔW with time of aluminum immersed in different concentrations of extract in 0.6 M HCl at 303-323 K

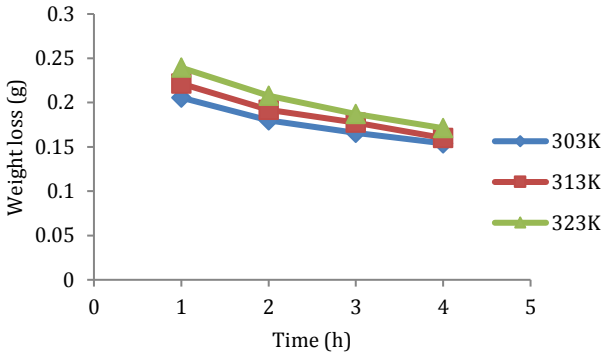


Figure 2: Variation of weight loss with time of the corrosion inhibition of aluminum with temperature at 0.6 HCl

Table 2: Adsorption isotherm for the adsorption of *DB* extract onto the Al surface in 0.6 M HCl

Isotherms	Temp. (K)	Parameters				
		Intercept	Slope	R^2	K_{ads}	ΔG (kJ mol^{-1})
Langmuir	303	1.5602	0.0029	1	344.83	-24.842
	313	1.7426	0.0013	1	769.23	-27.751
	323	2.207	0.0164	0.9994	60.98	-21.829
Temkin	303	0.0127	0.6426	0.6508	3.919	-90.66
	313	0.0016	0.5733	0.2795	6.300	-719.8
	323	0.0186	0.4515	0.9213	1.883	-61.91

The adsorption data show that the interaction of *DB* extract with the aluminum surface is best described by the Langmuir isotherm. This is evident from the very high correlation coefficients ($R^2 \approx 1$ at 303 K and 313 K, and 0.9994 at 323 K), indicating a strong linear fit. This suggests monolayer adsorption on a homogeneous aluminum surface with no interaction between adsorbed species [1]. The K_{ads} is relatively high, particularly at 313 K (769.23), indicating strong adsorption affinity between inhibitor molecules and the aluminum surface. However, K_{ads} decreases at 323 K (60.98), implying that adsorption weakens at higher temperatures due to thermal desorption of inhibitor molecules. The ΔG values for Langmuir are negative (−21.829 to −27.751 kJ mol^{-1}), confirming that the adsorption process is spontaneous [19]. The magnitude (between −20 and −40 kJ mol^{-1}) suggests that the mechanism involves both physisorption and weak chemisorption, though predominantly physical adsorption [1]. In contrast, the Temkin isotherm shows poor correlation (low R^2 values, especially at 313 K), indicating it does not adequately describe the system (Table 2). The inconsistent ΔG values further support that it is not a fit for the corrosion process in corrosion inhibition. This result is similar to a result on the aluminum surface obtained by [20].

The energy parameters in Table 3 provide insight into the mechanism of aluminum corrosion and the effect of *DB* extract in 0.6 M HCl. The activation energy (ΔE_a) increases from 53.75 kJ mol^{-1} (blank) to 58.22 kJ mol^{-1} at 0.6 g extract concentration. This rise indicates that the presence of the extract raises the energy barrier for corrosion, thereby slowing the dissolution of aluminum [21]. This behavior is typical of physical adsorption, where inhibitor molecules form a protective barrier that hinders access of corrosive ions to the metal surface [22]. Similarly, the enthalpy of activation (ΔH_{ads}) shows a slight increase with concentration (0.231 to 0.249 kJ mol^{-1}). The positive values confirm that the corrosion process is endothermic, meaning higher temperatures favor the reaction [10,20,23]. However, the increase in ΔH_{ads} in the presence of the inhibitor suggests that more energy is required for metal dissolution, further supporting the protective action of the extract. The entropy of activation

(ΔS) values are negative and nearly constant (≈ −0.197 kJ mol^{-1}), indicating that the activated complex is more ordered than the reactants. This result is similar to the results obtained by Iorhuna *et al*. [10] which implies a decrease in randomness during the formation of the transition state, often associated with the adsorption of inhibitor molecules organizing at the metal-solution interface.

Table 4 presents the kinetic parameters for aluminum corrosion in 0.6 M HCl in the absence and presence of *DB* extract across different temperatures. At 303 K, the rate constant (k_1) decreases slightly with increasing extract concentration (1.238 to 1.181), while the half-life ($t_{1/2}$) increases (0.560 to 0.587). This indicates that the corrosion reaction becomes slower in the presence of the inhibitor, and the metal takes longer to dissolve [23]. The high R^2 values (≈0.98–0.99) confirm a good fit to the kinetic model, suggesting reliable data. The increase in half-life supports the formation of a protective adsorbed layer that delays corrosion. At 313 K, k_1 is generally higher than at 303 K, indicating that CR increases with temperature, as expected for thermally activated processes. However, increasing inhibitor concentration still reduces k_1 (1.386 to 1.298) and increases $t_{1/2}$ (0.500 to 0.534). This shows that the extract retains inhibitory action, although less effective than at lower temperatures [24]. The lower and inconsistent R^2 values at intermediate concentrations suggest some instability in adsorption at this temperature. At 323 K, k_1 values are highest overall, confirming that corrosion is fastest at elevated temperatures [25]. Interestingly, k_1 increases with concentration at some points, and $t_{1/2}$ slightly decreases, indicating reduced %IE. This is due to desorption of inhibitor molecules from the aluminum surface at high temperatures according to Rout *et al*. [26]. However, the high R^2 at 0.6 g (0.999) suggests better stability at higher concentration.

Table 3: Energy parameters for the dissociation of Al in HCl in the absence and presence of *DB*

Corrodent	Extract conc. (g/L)	ΔE_a (kJ mol^{-1})	ΔH_a (kJ mol^{-1})	ΔS (kJ mol^{-1})
0.6 M HCl	Blank	53.750	0.231	-0.197
	0.2	56.260	0.239	-0.198
	0.4	57.590	0.245	-0.197
	0.6	58.220	0.249	-0.197

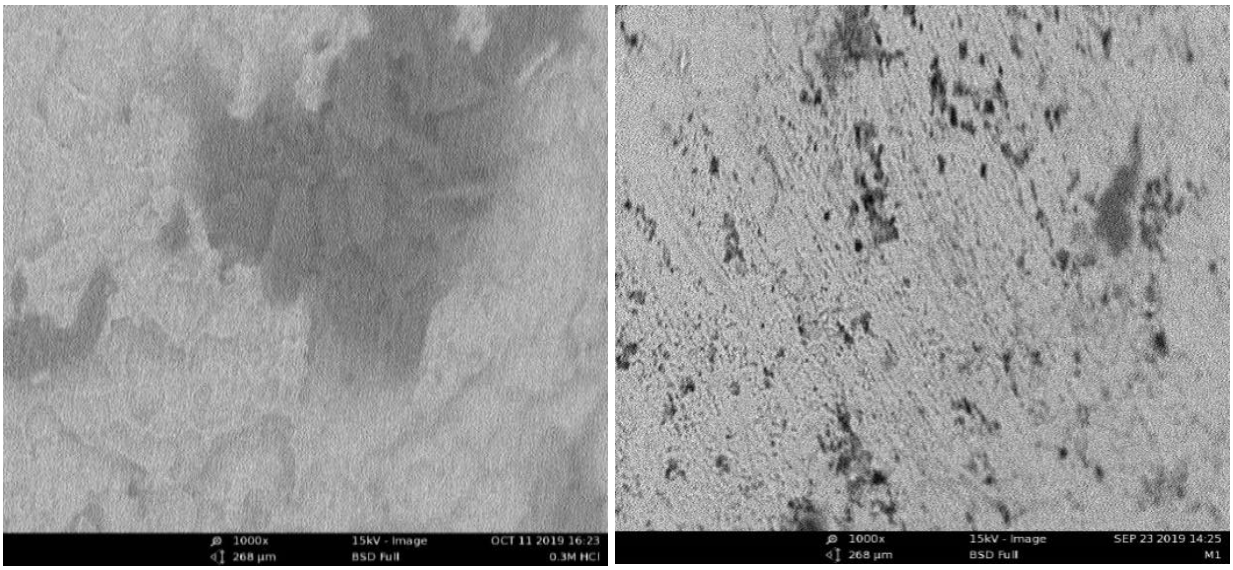
Table 4: Kinetic parameters for the corrosion of aluminum in 0.6 M HCl containing various concentrations of *DB* varying temperatures

Temp. (K)	Conc. (g/L)	0.6 M		
		k_1	$t_{1/2}$	R^2
303	0.0	1.238	0.560	0.991
	0.2	1.224	0.566	0.982
	0.4	1.222	0.567	0.987
	0.6	1.181	0.587	0.990
313	0.0	1.386	0.500	0.876
	0.2	1.361	0.509	0.765
	0.4	1.354	0.512	0.698
	0.6	1.298	0.534	0.987
323	0.0	1.392	0.498	0.943
	0.2	1.420	0.487	0.912
	0.4	1.450	0.475	0.654
	0.6	1.438	0.482	0.999

In Figure 3, upon immersion in 0.6 M HCl containing *DB* leaf extract, the aluminum surface exhibits noticeable roughness and scattered dark regions, indicating the occurrence of partial pitting corrosion under a more aggressive acidic environment. These pits are irregularly

distributed, reflecting localized acid attack that is only partially suppressed by the inhibitor. The increased acid concentration intensifies metal dissolution; however, the presence of the extract mitigates this effect to some extent. Phytochemical constituents such as polyphenols and flavonoids adsorb onto the aluminum surface, forming a protective but not fully continuous film 22. This adsorbed layer reduces the active sites available for corrosion, thereby lowering the rate of degradation [24]. The presence of clustered deposits suggests a combination of corrosion products and adsorbed inhibitor molecules, indicating competitive interactions at the metal solution interface. Overall, the extract demonstrates moderate %IE in 0.6 M HCl, slowing down corrosion processes and providing partial surface protection, although its effectiveness is reduced compared to less concentrated acidic conditions due to the increased aggressiveness of the medium.

Figure 3 shows the micrograph before and after corrosion inhibition.



SEM result before inhibition 0.6 M HCl

SEM after inhibition at 0.6 M HCl

Figure 3: SEM for inhibited aluminum sample in 0.6 M HCl using *DB* extract

Conclusion

The ΔW results clearly demonstrate that *DB* extract is an effective corrosion inhibitor for aluminum in 0.6 M HCl. The decrease in ΔW and CR with increasing extract concentration at all temperatures confirms that the inhibitor forms a protective film on the aluminum surface. This is further supported by the increase in surface coverage and %IE, indicating enhanced adsorption of inhibitor molecules. However, temperature significantly affects performance. As temperature increases from 303 to 323 K, CR increases while %IE decreases, even at higher concentrations. This suggests that the protective film becomes less stable at elevated temperatures due to desorption of inhibitor molecules from the metal surface. The overall %IE, reaching about 63–64% at 303 K, shows moderate effectiveness of the extract. The observed behavior indicates that the adsorption mechanism is predominantly physical, involving weak intermolecular interactions. Therefore, while the extract provides appreciable protection at lower temperatures, its efficiency diminishes at higher temperatures, highlighting temperature sensitivity in the corrosion inhibition process.

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